PES UNIVERSITY ONLINE

OPERATING SYSTEMS

Process Coordination & Deadlocks

S.Thenmozhi

Department of Computer Applications

Deadlock Recovery

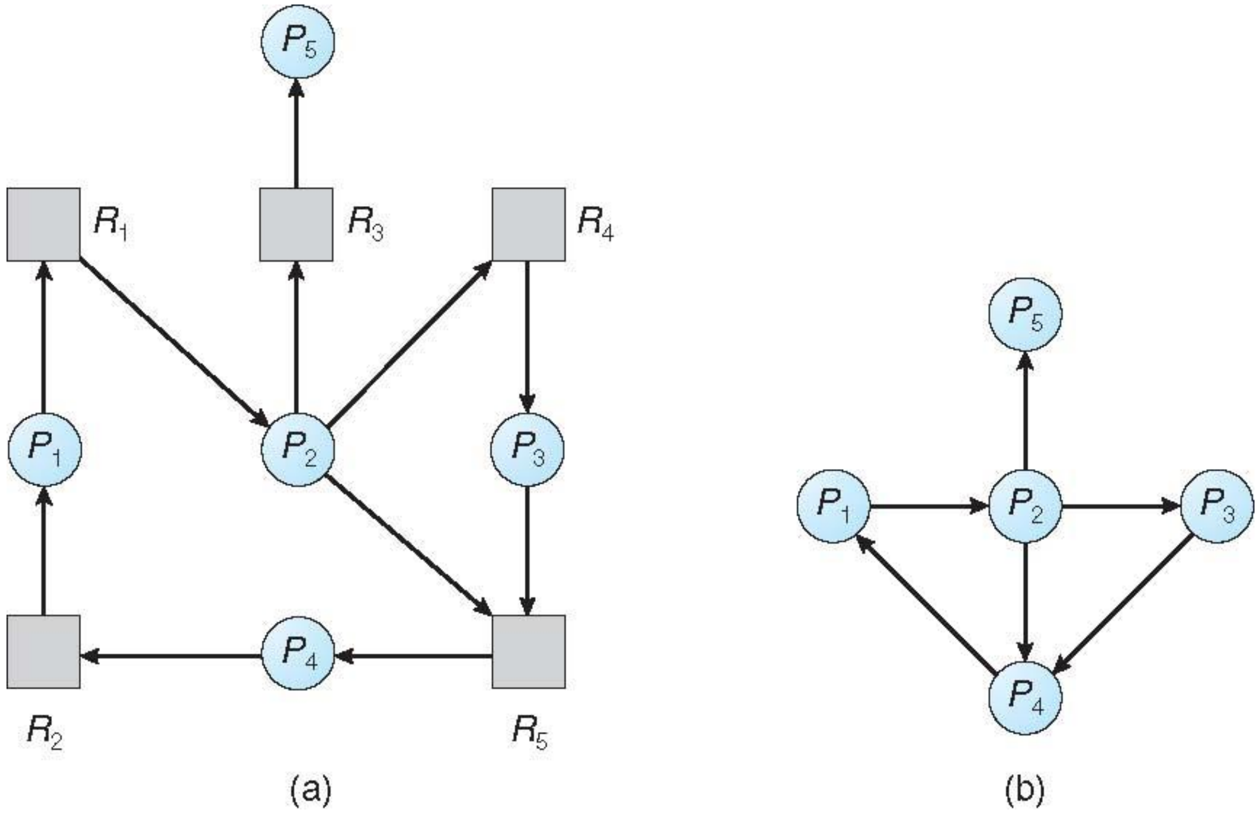
- Allow system to enter deadlock state
- Detect using algorithm
- Recovery scheme

Single Instance

- Maintain *wait-for* graph
  - Nodes are processes
  - $P_i \rightarrow P_j$ if P_i is waiting for P_j

- Periodically invoke an algorithm that searches for a cycle in the graph. If there is a cycle, there exists a deadlock

Deadlock Recovery



Resource-Allocation Graph   Corresponding wait-for graph

Multiple Instance: Apply bankers Algorithm to detect

Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	0	0	0	0	0	0
P1	2	0	0	2	0	2			
P2	3	0	0	0	0	1			
P3	2	1	1	1	0	0			
P4	0	0	2	0	0	2			

Deadlock exists, consisting of processes P_1, P_2, P_3, and P_4

Process Termination

- Abort all deadlocked processes
- Abort one process at a time until the deadlock cycle is eliminated
- How to choose the process for termination?
  - Priority of the process
  - How long process has computed, and how much longer to completion
  - Resources, the process has used
  - Resources required for the process to complete
  - How many processes will need to be terminated
  - Is process interactive or batch?

Resource Preemption

- Selecting a victim – minimize cost
- Rollback – return to some safe state, restart process for that state
- Starvation – same process may always be picked as victim, include number of rollback in cost factor

THANK YOU

S. Thenmozhi
Department of Computer Applications
thenmozhis@pes.edu
+91 80 6666 3333 Extn 393